RCS04 | Student Packet

CTS Cohort 6 | CTS Day 12 | 2026-07-23 | 10:00 AM-3:00 PM | 6 students

Plumbing Basics: Supply, Drainage, and Dry Verification

Core rule: A professional does not guess. A professional checks, protects, and verifies.

Big idea: Water follows pressure and gravity; reliable plumbing depends on correct parts, alignment, slope, seals, access, and verification.

Learning targets

- Differentiate supply and drain-waste-vent functions
- Explain fixture, trap, and shutoff roles
- Plan a training plumbing path through framing
- Assemble and troubleshoot a dry rolling-sink/P-trap module

Today's run

10:00-10:15 | Arrival, PPE, retrieval warm-up

10:15-10:30 | Morning quiz

10:30-10:50 | Quiz review and system-path check

10:50-11:20 | Supply and DWV system mapping

11:20-12:00 | P-trap parts, washer, and alignment demonstration

12:00-12:30 | Dry-verification guided practice and lab release

12:30-1:00 | Lunch and controlled reset

1:00-2:25 | Rolling-sink module rebuild and capstone pathway lab

2:25-2:45 | Fault-find, dry recheck, evidence, cleanup

2:45-3:00 | Exit ticket and Day 5 preview

Key words

supply | shutoff | fixture | tailpiece | P-trap | trap arm | DWV | slope | washer | slip nut | cleanout

Guided worksheet

1. Draw the supply and drain paths.

_____

_____

2. Label the P-trap, tailpiece, trap arm, and slip-joint washer.

_____

_____

3. Explain why cross-threading causes failure.

_____

_____

4. Mark a safe training path that avoids the electrical boxes.

_____

_____

5. Write the dry-verification sequence.

_____

_____

Afternoon lab mission

Teams rebuild and troubleshoot rolling sink/P-trap modules using dry verification only, then each student maps a coordinated plumbing pathway on the capstone wall without making unapproved structural penetrations.

Safety boundary

- No water or pressure testing unless the instructor explicitly authorizes and controls it
- No electrical tools or cords in the plumbing station
- Wipe any unexpected moisture immediately
- No structural penetration without instructor approval
- Stop for cracked, missing, or cross-threaded parts

Build / test sequence

[ ] Inventory module

[ ] Lay out system path

[ ] Inspect parts

[ ] Dry-fit

[ ] Orient washers

[ ] Hand assemble

[ ] Alignment inspection

[ ] Instructor inserts or identifies one fault

[ ] Diagnose one issue at a time

[ ] Correct and dry-reverify

[ ] Map capstone path

[ ] Document and reset

Evidence record

[ ] System map: _____

[ ] P-trap parts identification: _____

[ ] Dry assembly inspection: _____

[ ] Troubleshooting record: _____

[ ] Coordinated capstone pathway sketch: _____

Reflection and exit ticket

Draw the drain path

_____

_____

Name one assembly fault

_____

_____

Describe the dry-verification boundary

_____

_____

Explain one framing/plumbing coordination choice

_____

_____

Instructor verification

Status: [ ] Approved [ ] Correct and recheck [ ] Needs follow-up

Instructor initials: _____ Date/time: _____

RCS04 | Morning Quiz

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Plumbing Basics: Supply, Drainage, and Dry Verification

Name: _____ Date: _____

Answer from your own understanding. This is a teaching check, not a trap.

1. What system delivers water to a fixture?

_____

_____

2. What does DWV stand for?

_____

_____

3. What is the P-trap's basic purpose?

_____

_____

4. What should happen before tightening slip joints?

_____

_____

5. Why troubleshoot one change at a time?

_____

_____

6. What does a shutoff valve allow?

_____

_____

7. May students drill framing for plumbing without approval?

_____

_____

8. What type of verification is planned for today's lab?

_____

_____

9. Why keep plumbing and electrical stations separated?

_____

_____

10. What does a professional do when a part is cracked or cross-threaded?

_____

_____